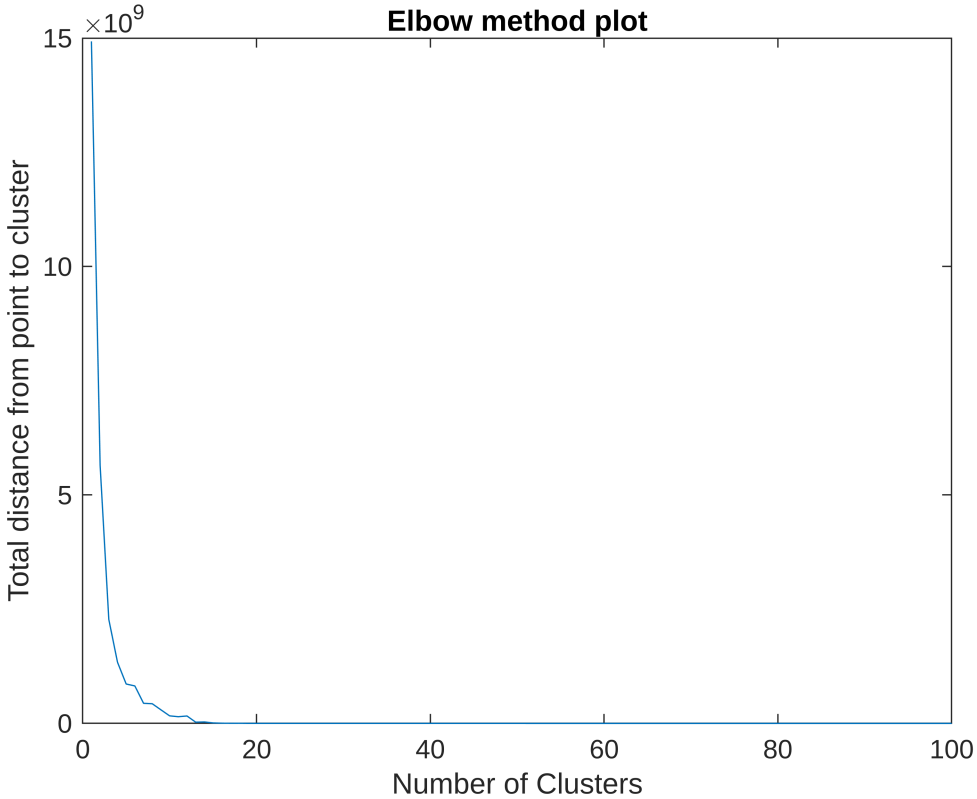
download of the dataset and verification of no correlation between the different features through a scatter plot of every feature with the others (could have done a confusion matrix too)

```
clear,close all;
data=load("airfoil_self_noise.dat");
% for i=1:5
%     for j=2:6
%         figure;
%         scatter(data(:,i),data(:,j));
%         title([num2str(i) "and" num2str(j)]);
%     end
% end
```

let's see for clustering, here the elbow method, we can see the stagnation at 13 clusters

```
totaldistance=zeros(1,100);
for i=1:100
    [Kid,Kcenters,Kdsum]=kmeans(data,i);
    % Kdsum=Kdsum/max(Kdsum);
    for k=1:length(Kdsum)
        totaldistance(1,i)=totaldistance(1,i)+Kdsum(k);
    end
end
figure;
plot(totaldistance);
title('Elbow method plot');
xlabel('Number of Clusters');
ylabel('Total distance from point to cluster')
```




```
max(data(:,1))
```

```
ans =
20000
```

```
min(data(:,1))
```

```
ans =
200
```

```
max(data(:,2))
```

```
ans =
22.2000
```

```
min(data(:,2))
```

```
ans =
0
```

```
max(data(:,3))
```

```
ans =
0.3048
```

```
min(data(:,3))
```

```
ans =
0.0254
```

```
max(data(:,4))
```

```
ans =
71.3000
```

```
min(data(:,4))
```

```
ans =
31.7000
```

```
max(data(:,5))
```

```
ans =
0.0584
```

```
min(data(:,5))
```

```
ans =
4.0068e-04
```

```
% for i=1:100
%     [c,u,obj]=fcm(data,i);
%     % Kdsum=Kdsum/max(Kdsum);
%     cost(i)=min(obj);
% end
% figure;
% plot(cost);
% title('Elbow method plot');
% xlabel('Number of Clusters');
% ylabel('Total distance from point to cluster')
```

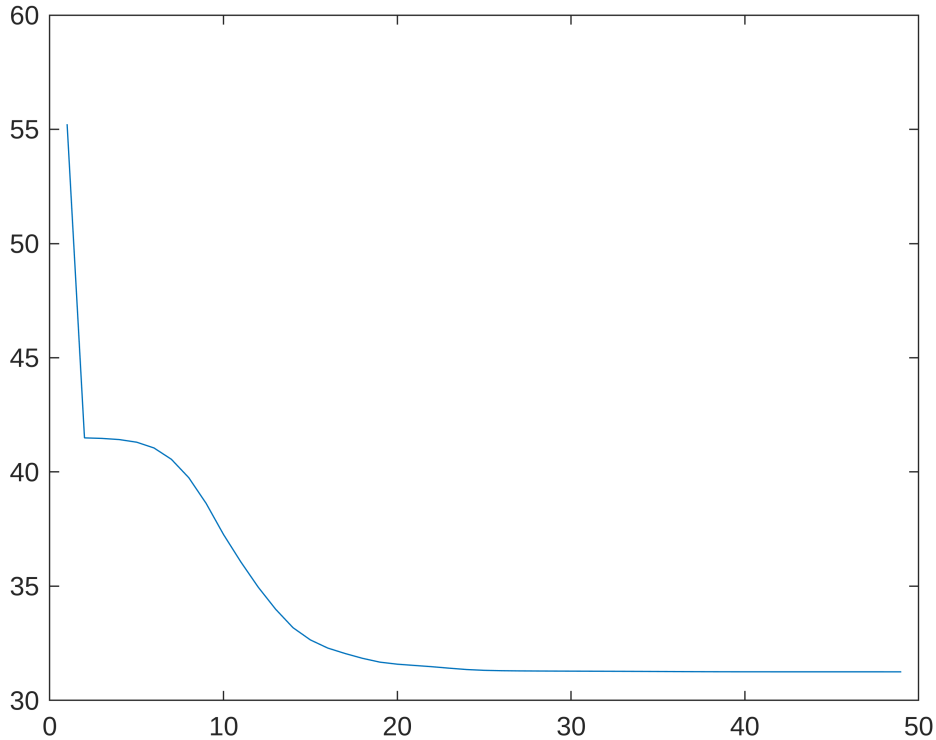
Normalization of the dataset for smoother processing

```
for input=1:5
    data(:,input)=(data(:,input)-min(data(:,input)))/(max(data(:,input))-
min(data(:,input)));
end
train=randperm(1503,ceil(1503*0.8));
points=[];
for p=1:length(data)
    if isempty(find(train==p))
        points=[points p];
    end
end

datatrain=data(train,:);
datatest=data(points,:);

% fcm clustering
[clusters, U,fobj] = fcm(data(:, 1:end-1), 15);  % run a fcm clustering
Exclude output column
```

```
Iteration count = 1, obj. fcn = 55.2258
Iteration count = 2, obj. fcn = 41.4927
```

```
Iteration count = 3, obj. fcn = 41.4699
Iteration count = 4, obj. fcn = 41.4192
Iteration count = 5, obj. fcn = 41.3039
Iteration count = 6, obj. fcn = 41.0509
Iteration count = 7, obj. fcn = 40.5555
Iteration count = 8, obj. fcn = 39.7512
Iteration count = 9, obj. fcn = 38.6246
Iteration count = 10, obj. fcn = 37.2631
Iteration count = 11, obj. fcn = 36.0592
Iteration count = 12, obj. fcn = 34.9545
Iteration count = 13, obj. fcn = 33.9908
Iteration count = 14, obj. fcn = 33.1827
Iteration count = 15, obj. fcn = 32.6477
Iteration count = 16, obj. fcn = 32.2912
Iteration count = 17, obj. fcn = 32.0522
Iteration count = 18, obj. fcn = 31.8417
Iteration count = 19, obj. fcn = 31.675
Iteration count = 20, obj. fcn = 31.5857
Iteration count = 21, obj. fcn = 31.5299
Iteration count = 22, obj. fcn = 31.4747
Iteration count = 23, obj. fcn = 31.41
Iteration count = 24, obj. fcn = 31.3511
Iteration count = 25, obj. fcn = 31.3168
Iteration count = 26, obj. fcn = 31.3005
Iteration count = 27, obj. fcn = 31.2921
Iteration count = 28, obj. fcn = 31.287
Iteration count = 29, obj. fcn = 31.2836
Iteration count = 30, obj. fcn = 31.2809
Iteration count = 31, obj. fcn = 31.2784
Iteration count = 32, obj. fcn = 31.2758
Iteration count = 33, obj. fcn = 31.2729
Iteration count = 34, obj. fcn = 31.2695
Iteration count = 35, obj. fcn = 31.2658
Iteration count = 36, obj. fcn = 31.262
Iteration count = 37, obj. fcn = 31.2586
Iteration count = 38, obj. fcn = 31.2559
Iteration count = 39, obj. fcn = 31.254
Iteration count = 40, obj. fcn = 31.2529
Iteration count = 41, obj. fcn = 31.2522
Iteration count = 42, obj. fcn = 31.2519
Iteration count = 43, obj. fcn = 31.2517
Iteration count = 44, obj. fcn = 31.2516
Iteration count = 45, obj. fcn = 31.2515
Iteration count = 46, obj. fcn = 31.2515
Iteration count = 47, obj. fcn = 31.2515
Iteration count = 48, obj. fcn = 31.2515
Iteration count = 49, obj. fcn = 31.2515
Minimum improvement reached.
```

```
figure;
plot(fobj);
```



Now let's begin the inference system and problem definition

We have a 5 inputs 1 output FIs that will be trained using a cascading GA and then for the inference, we will use 2 systems, a TSK 1st order and a TSK 0 order to compare efficiency.

```
% GA parameters
% stochastic rates :
selection_rate=0.01;
elitism_rate=0.001;
mutation_rate=0.3;

% generational parameters
Popsize=200;
maxgen=500;
```

Let's initialize both the TSKs

```
%TSK0_pop=initTSK0(Popsize);
cluster_pop=initfcm(Popsize,clusters);
% TSK1_pop=initTSK1(Popsize);
% TSK1_pop=TSK1_pop';
%TSK0_pop=TSK0_pop';
parfor i=1:Popsize
    fit0(i)=fitfcm(cluster_pop{i},datatrain,clusters);
end
```

now let's do the optimization

```
bestfit=[];
for gen=1:maxgen

    mutation_rate=max(0.5,1-gen/maxgen);
    [num_elites,pool] = selection(cluster_pop, fit0,
selection_rate,Popsize,elitism_rate);
    cluster_pop = reproduction(pool, mutation_rate,
Popsize,num_elites,clusters);

    % now the new fitness
    bestfit(gen)=min(fit0); % store best fitness
    parfor i=1:Popsize
        newfit(i)=fitfcm(cluster_pop{i},datatrain,clusters);
    end
    fit0=newfit;
    disp(["gen" gen ", best fit = " bestfit(gen) "avg=" bestfit(gen)/
length(datatrain)])
end
```

```
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```
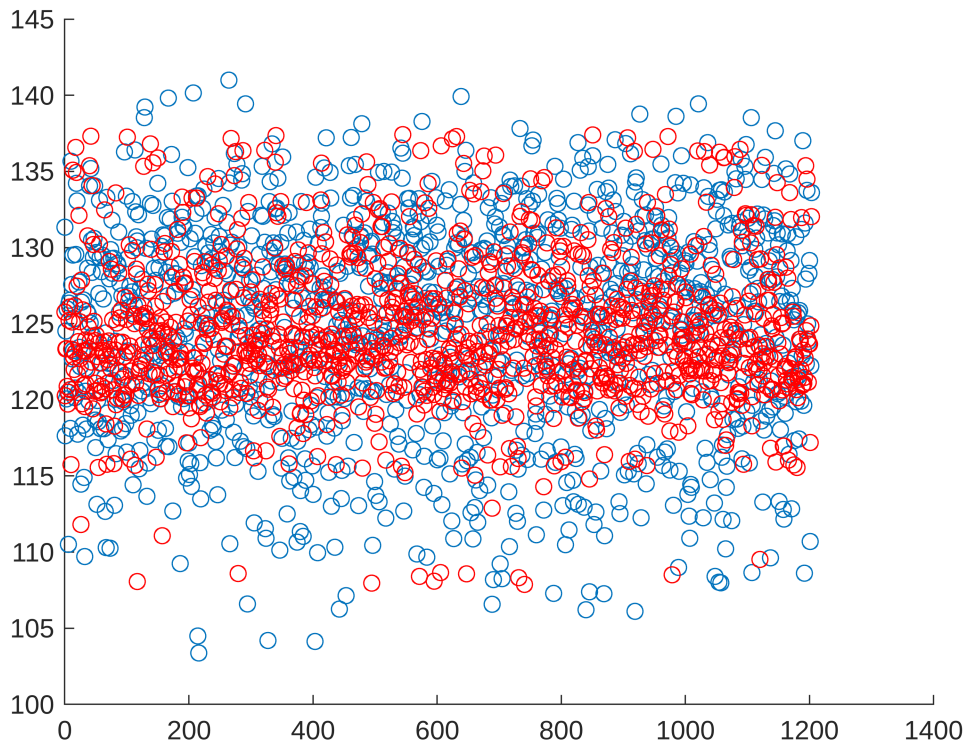
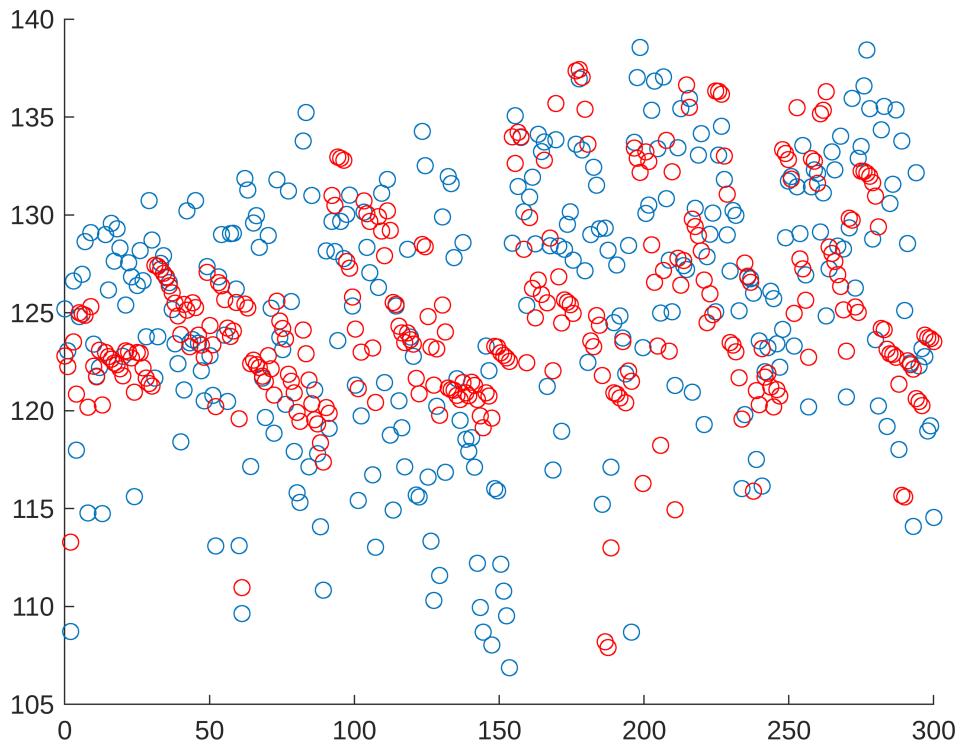

```
[~, sorted_idx] = sort(fit0, 'ascend');
out=visucluster(cluster_pop{sorted_idx(1)},datatrain,clusters)
```



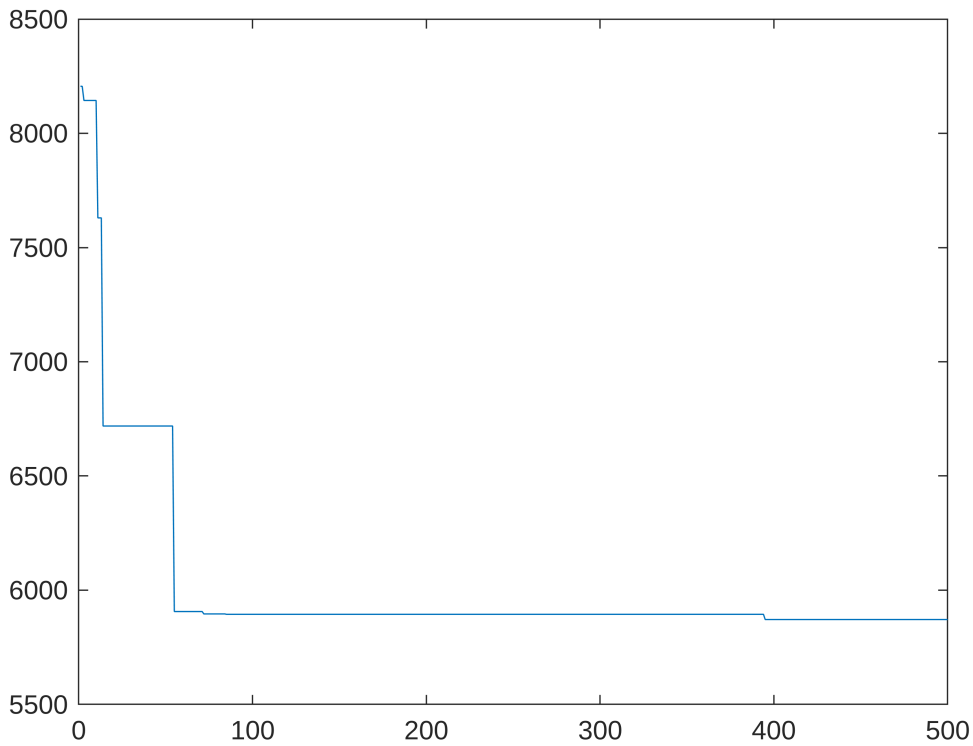

```
out =
5.8705e+03
```

```
[~, sorted_idx] = sort(fit0, 'ascend');
out=visucluster(cluster_pop{sorted_idx(1)},datatest,clusters)
```




```
out =
1.4206e+03
```

```
figure;
plot(bestfit)
```



here are all the functions used in the .mlx

```
% TSK initializations
%  real mf bounds : 0:20000; 0:25; 0:0.35; 0:75; 0:0.06;
% syntetic mf : 0:1; 0:1; 0:1 /normalized value
function pop=initfcm(popsize,clusters)
    pop=cell(popsize,1);

    mfs=[];
    for i=1:popsize
        for k=1:max(size(clusters))
            rule_param=50*(rand(1,6)-0.5); % parameters between -10 and 10;
            rule_param(1,6)=120+100*(rand(1,1)-0.75); % constant between 70
and 170
            pop{i}=[pop{i}; mfs rule_param];
        end
    end


end





function pop=initcluster(popsize,clusters)
```

```
    pop=cell(popsize,1);
    % we will use a gaussian membership function based on the different
    % cluster centers and the optimization parameter will be the standard
    % deviation "std"

    % we then have a total of 5 std to tune plus 6 rule parameter per
    % cluster, totaling 11*13=143 parameters

    mfs=[];
    for i=1:popsize
        for k=1:max(size(clusters))
            mfs=0.05+0.15*rand(1,5); % initialize the std the between 0.05
and 0.1  for each cluster
            rule_param=20*(rand(1,6)-0.5); % parameters between -5 and 5;
            rule_param(1,6)=120+110*(rand(1,1)-0.5);
            pop{i}=[pop{i}; mfs rule_param];
        end
    end
end


%selection, mutation, reproduction
function [num_elites,pool] = selection(pop, fitness_vals,
rate,popsize,e_rate)
    % popsize = length(pop);
    num_selected = round(rate * popsize);
    num_elites = ceil(e_rate * popsize);
    [~, sorted_idx] = sort(fitness_vals, 'ascend');
    elites = pop(sorted_idx(1:num_elites));  % Preserve the best individuals
    if size(elites,1)~=1
        elites=elites';
    end
    % Tournament Selection (forces selection pressure)
    num_to_select = num_selected - num_elites;
    selected_individuals = cell(1, num_to_select);
    for i = 1:num_to_select
        % Pick 3 random individuals and select the best one
        candidates = randperm(popsize, 3);
        [~, best_idx] = min(fitness_vals(candidates));
        selected_individuals{i} = pop{candidates(best_idx)};
    end

    % Combine elites and selected individuals
    pool = [elites, selected_individuals];
end

function child = mutation(child, mutation_rate,clusters)
    if rand < mutation_rate
        if rand>0.66
            new_child=initfcm(1,clusters);
```

```
            child=new_child{1};
        else

            child=child.*(1+(0.4*rand(15,6)-0.5)); % up to +-20%
        end
    end
end

function new_pop = reproduction(pool, mutation_rate,
popsize,num_elites,clusters)
    new_pop = cell(1, popsize);

    for i = 1:num_elites
        new_pop{i} = pool{i}; % Directly copy elite individuals
    end
    for i = num_elites+1:popsize
        parent1 = pool{randi(length(pool))}; %select parents
        parent2 = pool{randi(length(pool))};
        child=zeros(15,6);
        % we want to do crossover on each membership function by doing an
        % interpolation of the parents parameters and crossover on rules
        alpha=rand(1);
        beta=1-alpha;
        child=parent1*alpha+parent2*beta;
        % child(1:13,1:5)=parent1(1:13,1:5)*alpha + parent2(1:13,1:5)*beta;
        % child(1:8,:)= [parent1(1:8,:)];
        % child(9:15,:)=[parent2(9:15,:)];

        % Mutation
        child = mutation(child, mutation_rate,clusters);

        new_pop{i} = child;
    end
end
function f=fitfcm(chr,data,ccs)

    for ind=1:length(data)
        output=0;
        for k = 1:size(ccs,2)
            dist_ik = norm(data(ind,1:5) - ccs(k,:)) + 1e-6; % small value
to avoid div by zero
            denom = 0;
            for j = 1:size(ccs,2)
                dist_ij = norm(data(ind,1:5) - ccs(j,:)) + 1e-6;
                denom = denom + (dist_ik / dist_ij)^(2/(2.5-1));
            end
            U_test(k) = 1 / denom;

            a = chr(k,1:5); % 5 coefficients
            b = chr(k,6);   % bias
```

```
            y_k = dot(a, data(ind,1:5)) + b;
            output = output + U_test(k) * y_k;
        end
        pred(ind,1) = output;
    end
    errors = abs(pred - data(:,6));
    f = sum(errors)+10*max(errors);


end

function mu=fcmeval(datatest, clusters, m)
    c = size(clusters, 1); % number of clusters
    n = size(datatest, 1); % number of data points
    U_test = zeros(c, n);
    output=0;
    for i = 1:n
        for k = 1:c
            dist_ik = norm(datatest(i,:) - clusters(k,:)) + 1e-6; % small
value to avoid div by zero
            denom = 0;
            for j = 1:c
                dist_ij = norm(datatest(i,:) - clusters(j,:)) + 1e-6;
                denom = denom + (dist_ik / dist_ij)^(2/(m-1));
            end
            U_test(k,i) = 1 / denom;
        end
    end
end

function f=fitcm(chr,data,ccs)
   for ind=1:length(data)
       w=ones(max(size(ccs)),1);
        % let's compute the firing power of each rule
        for k=1:max(size(ccs))
            for input=1:5
                w(k)=w(k)*exp(-(data(ind,input)-ccs(k,input))^2/
(2*chr(k,input)^2)); % based on gaussian membership
            end
        end
        agregg=sum(w);
        output = 0;
        for k = 1:max(size(ccs))
            x_inputs = data(ind, 1:5);
            rule_params = chr(k, 6:11); % [a1...a5, b]
            rule_val = dot(rule_params(1:5), x_inputs) + rule_params(6);
            output = output + w(k) * rule_val;
        end
        pred(ind)=output/(agregg+10^(-10));
   end
```

```
   errors = abs(pred' - data(:,6));
   f = sum(errors);
end
% fitness function for 0 order tsk

function out=visu(chr, data,ccs)
    for ind=1:length(data)
       w=ones(max(size(ccs)),1);
        % let's compute the firing power of each rule
        for k=1:max(size(ccs))
            for input=1:5
                w(k)=w(k)*exp(-(data(ind,input)-ccs(k,input))^2/
(2*chr(k,input)^2)); % based on gaussian membership
            end
        end
        agregg=sum(w);
        output = 0;
        for k = 1:max(size(ccs))
            x_inputs = data(ind, 1:5);
            rule_params = chr(k, 6:11); % [a1...a5, b]
            rule_val = dot(rule_params(1:5), x_inputs) + rule_params(6);
            output = output + w(k) * rule_val;
        end
        pred(ind)=output/(agregg+10^(-10));
   end
   errors = abs(pred' - data(:,6));
   out = sum(errors);

    figure;
    scatter(linspace(0,length(data(:,6)),length(data(:,6))),data(:,6))
    hold on
    scatter(linspace(0,length(data(:,6)),length(data(:,6))),pred,"red")

    % out=output;
end

function f=visucluster(chr,data,ccs)

    for ind=1:length(data)
        output=0;
        for k = 1:size(ccs,2)
            dist_ik = norm(data(ind,1:5) - ccs(k,:)) + 1e-6; % small value
to avoid div by zero
            denom = 0;
            for j = 1:size(ccs,2)
                dist_ij = norm(data(ind,1:5) - ccs(j,:)) + 1e-6;
                denom = denom + (dist_ik / dist_ij)^(2/(2-1));
            end
            U_test(k) = 1 / denom;
```

```
            a = chr(k,1:5); % 5 coefficients
            b = chr(k,6);   % bias
            y_k = dot(a, data(ind,1:5)) + b;
            output = output + U_test(k) * y_k;
        end
        pred(ind,1) = output;
    end
    errors = abs(pred - data(:,6));
    f = sum(errors);

    figure;
    scatter(linspace(0,length(data(:,6)),length(data(:,6))),data(:,6))
    hold on
    scatter(linspace(0,length(data(:,6)),length(data(:,6))),pred,"red")
end
```